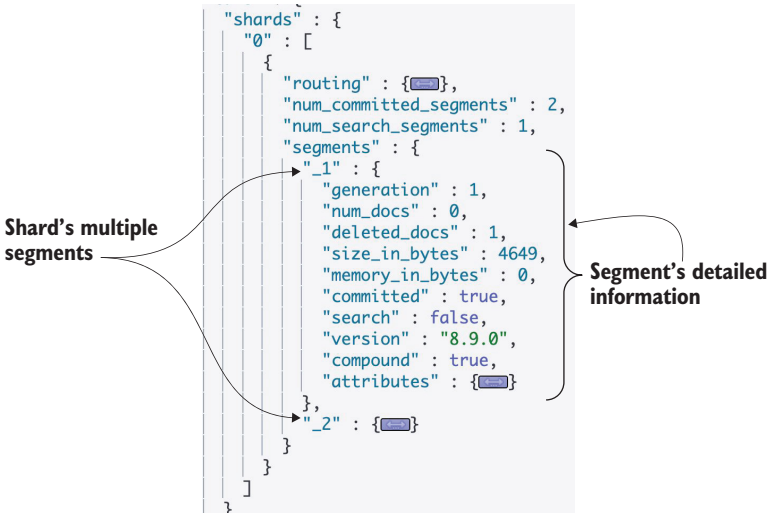


Figure 6.9 Detailed information about every segment in a shard

Sometimes we may need to manage an overgrown index to save space or reduce the infrastructure behind a predominantly dormant index. We can perform such operations using advanced tricks, discussed in detail in the next section.

6.8 Advanced operations

Earlier, we looked at CRUD operations such as creating, reading, and deleting indexes. In addition to such basic operations, we can perform advanced operations like splitting an index to add more shards or downsizing an index by shrinking it or rolling it over periodically (such as daily). Let's review a few advanced operations we can perform on our indices.

6.8.1 Splitting an index

Sometimes indexes are overloaded with data. To avoid the risk of losing data or mitigate slower search query responses, we can redistribute the data onto additional shards. Adding shards to the index optimizes memory and distributes documents evenly.

For example, if an index (`cars`) with 5 primary shards is overloaded, we can split the index into a new index with more primary shards: say, 15. Expanding an index from a small size to a larger size is called *splitting* the index. Splitting is nothing more than creating a new index with more shards and copying the data from the old index into the new index.

Elasticsearch provides a `_split` API for splitting an index. There are some rules, such as how many shards a new index can be created with, but first let's see how we can split an index.